

DBMS Project Assignment-1: Requirement Specification

Project Title

QuickShow – Online Movie Ticket Booking and Theatre Management System

Client

Galaxy Cinemas

Requirement Specification

Galaxy Cinemas is a cinema chain that operates multiple theatres across different cities. The management intends to replace its existing manual ticket booking process with a centralized web-based movie ticket booking and theatre management system. The proposed system should enable customers to browse movies, check show timings, select seats and book tickets online while allowing theatre administrators to efficiently manage movies, theatres, screens and bookings.

Customer Information

The system needs to keep track of every customer who registers. Each customer gets a unique User ID. Since names aren't really a single piece of data, Name is stored as a composite attribute made up of First Name, Middle Name and Last Name. Address works the same way: it's broken down into House Number, Street, Area, City, State and PIN Code.

Apart from this, the system also stores each customer's Phone Number, Date of Birth, Gender, Password, and Email Addresses, since a customer can register more than one email if needed. Age isn't stored directly; it's a derived attribute, calculated on the fly from the Date of Birth whenever it's actually needed.

Movie Information

Every movie in the system has a unique Movie ID. Along with the ID, the system stores the Movie Title, Poster, Language, Genre, Duration, Release Date, IMDb Rating, Certificate, Director, Cast and a short Description. Cast is treated as a multivalued attribute because a single movie can have several actors attached to it.

Theatre Information

Each theatre is identified by a unique Theatre ID, along with its Name, Contact Number and Address. Like the customer's address, the theatre's address is also composite: Building Name, Street, Area, City, State and PIN Code. A single theatre can have multiple screens under it.

Screen Information

Every screen inside a theatre has its own unique Screen ID, a Screen Name/Number, a Screen Type (2D, 3D, IMAX or 4DX), a Seating Capacity, and is linked to the theatre it belongs to.

Seat Information

Each seat is uniquely identified by a Seat ID. The seat's position is stored as a composite attribute combining Row Number and Seat Number. Every seat belongs to exactly one screen, and it can't exist across two screens at once.

Show Information

Every show scheduled in the system gets a unique Show ID and is linked to exactly one movie and one screen. Along with this, the system records the Show Date, Show Time, Movie ID, Screen ID, Ticket Price, Available Seats and the current Show Status. A single movie can be scheduled for multiple shows: different screens, different theatres, different dates.

Booking Information

Every booking made by a customer is tracked using a unique Booking ID. This record holds the customer who booked, the movie, theatre, screen and show involved, along with the booking date and time, the seats booked, the total seat count, booking status and the total amount payable.

Since one booking can cover multiple seats, Booked Seats is kept as a multivalued attribute.

Payment Information

Payments are tracked separately, each with a unique Payment ID. This includes the associated Booking ID, Payment Method, Transaction ID, Payment Amount, Payment Date & Time, and Payment Status.

Non-Functional Requirements

The interface should be simple and intuitive enough that both customers and admins can use it without difficulty. Data integrity and consistency need to be maintained at all times, and the system should handle multiple users booking seats at the same time without running into conflicts, with no duplicate seat allocations. Search and retrieval of movie data should be fast, customer credentials need to be stored securely, and overall, the system should be scalable enough to handle multiple theatres and a growing user base, while staying reliable.

Functional Requirements

Registered customers can log in with their email and password. Once logged in, they're able to:

- Browse all available movies
- Search for a movie by title
- View full movie details
- Filter movies by language, genre or certification
- Check which theatres are showing a movie, along with showtimes
- Select seats and book (or cancel) tickets

Whenever a booking gets confirmed, the seats belonging to that show are automatically marked unavailable, and the system makes sure the same seat can never be double-booked for the same show. And if a booking gets cancelled, those seats become available again.

On the admin side, the administrator manages the system, adding, updating or removing movies, managing theatres and screens, scheduling shows, updating ticket prices, seat availability, viewing customer bookings.